

HORNSBY GIRLS' HIGH SCHOOL



2008 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question on a fresh sheet of paper

Total marks (84)

- Attempt Questions 1 – 7
- All questions are of equal value

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Total Marks
Attempt Questions 1–7
All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your student number and question number at the top of the page. Extra paper is available.

Question 1 (12 marks) Use a SEPARATE sheet of paper. **Marks**

(a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$. **1**

(b) Find the acute angle between the lines $y = 2x - 9$ and $3y = x + 8$. **2**

(c) Find $\frac{d(3^x)}{dx}$. **1**

(d) State the domain and range of the function $y = 2 \cos^{-1} 3x$. **2**

(e) Use the substitution $u = \tan x$ to evaluate $\int_{\pi/6}^{\pi/4} \frac{dx}{\cos^2 x \tan x}$. **3**

(f) Consider the function $y = x \cos^{-1} x - \sqrt{1 - x^2}$,

(i) Show that $\frac{dy}{dx} = \cos^{-1} x$. **2**

(ii) Hence, or otherwise, evaluate $\int_0^1 \cos^{-1} x dx$. **1**

Question 2 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) Solve $x - 5 < \frac{14}{x}$.

3

(b) Find the general solution to $\tan 2\theta = \sqrt{3}$.
Express your answer in terms of π .

2

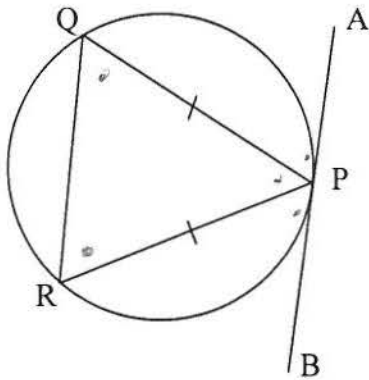
(c) The polynomial $f(x) = 2x^3 + ax^2 + bx + 6$ has a remainder of -6 when divided by $(x-1)$ and $f(-2) = 0$.
Find the values of a and b .

2

(d) Find the exact value of $\int_0^{\frac{\pi}{4}} \cos^2\left(\frac{1}{2}x\right) dx$.

3

(e)



Given that $PQ = PR$ and AB is a tangent to the circle PQR at P , prove that $RQ \parallel BA$.

2

Question 3 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) 8 people are to be seated at a round table
- (i) How many seating arrangements are possible? 1
 - (ii) Two people, Sarah and Ken, can not sit together.
How many seating arrangements are then possible? 2
- (b) The function $f(x) = x^3 + ax^2 + bx + c$ has a relative maximum at $x = \lambda$ and a relative minimum $x = \beta$.
- (i) Prove $\lambda + \beta = -\frac{2}{3}a$. 2
 - (ii) Show that a point of inflexion occurs at $x = \frac{\lambda + \beta}{2}$. 2
(A check for concavity is not required.)
- (c) A roast chicken has been taken from an oven and placed in a room of constant temperature 20°C . At time t minutes its temperature T decreases according to the equation
- $$\frac{dT}{dt} = -k(T - 20) \text{ where } k \text{ is a positive constant.}$$
- The initial temperature of the chicken is 80°C and cools to 50°C after 10 minutes.
- (i) Verify that $T = 20 + Ae^{-kt}$ is a solution of this equation where A is a constant. 1
 - (ii) Find the values of A and k . 2
 - (iii) How long will it take for the chicken to cool to 30°C ? 2
Give your answer to the nearest minute.

Question 4 (12 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) A body is in Simple Harmonic Motion and its position at a time t is given by the equation

$$x = R \cos(nt + \alpha) + 1.$$

The period of motion is π seconds. Initially the body is at rest 3 units to the left of the origin.

- (i) Find the values of R , n and α . 3
- (ii) Find the velocity of the body when $t = \frac{\pi}{6}$. 1
- (b) (i) Consider the equation $x \ln x - 1 = 0$. Show that a solution of this equation lies between $x = 1$ and $x = 2$. 1
- (ii) Using $x = 2$ as a first approximation for a solution, apply Newton's method once to find a better approximation. 2
Give your answer to 1 decimal place.
- (c) Prove the identity $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta$. 2
- (d) A 'Wheel of Chance' has 9 equal compartments around its rim.
When this wheel is spun a player can win \$100 on 1 designated compartment.
Grace is given the opportunity to have 25 consecutive spins of the wheel.
Find, giving your answer correct to 4 decimal places, the probability that she will win:
- (i) exactly \$200, 1
- (ii) at least \$200. 2

Question 5 (12 marks) Use a SEPARATE sheet of paper.

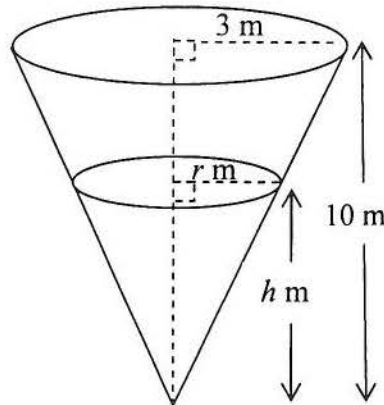
Marks

- (a) Use the principle of mathematical induction to show that

$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2 \quad \text{for all } n \geq 1$$

3

- (b)



The diagram shows a conical wheat flu. The flu is being filled with wheat at the rate of 2 m^3 per minute. The height of wheat at time t minutes is h metres and the radius of the wheat's top surface is r metres.

- (i) Show that $r = \frac{3h}{10}$.

1

- (ii) Find the rate at which the height is increasing when the height of the wheat is 8m.

3

$$(\text{Volume of cone} = \frac{1}{3}\pi r^2 h)$$

- (c) Solve $x^3 - 21x^2 + 126x - 216$ given that the roots form 3 consecutive terms of a geometric series.

3

- (d) Use Simpson's Rule with 3 function values to find an approximation to

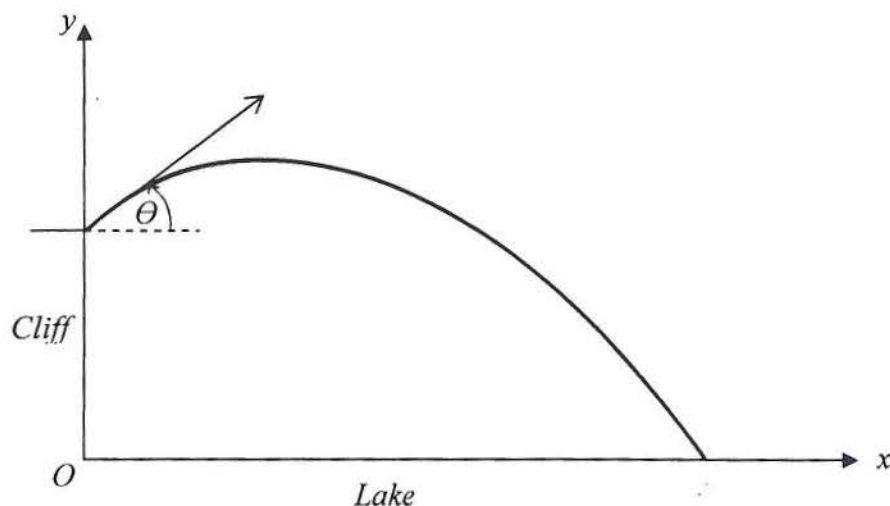
2

$$\int_0^{0.4} \sin^{-1} x \, dx \text{ to one decimal place.}$$

Question 6 (12 marks) Use a SEPARATE sheet of paper.

Marks

- (a) A stone is projected with a velocity of 10 metres per second at an angle of elevation of $\theta = \tan^{-1}\left(\frac{3}{4}\right)$ from the top of a cliff 27 metres high overlooking a lake.



Assume that the equations of motion of the stone are

$$\ddot{x} = 0 \quad \ddot{y} = -10$$

referred to the coordinate axes shown.

- (i) Let (x, y) be the position of the stone at time t seconds after it was thrown, and before the stone hits the lake. 2
It is known that $x = 8t$.
Show that $y = -5t^2 + 6t + 27$.
- (ii) Calculate the time which elapses before the stone hits the lake and find the horizontal distance of the point of contact from the base of the cliff. 3
- (iii) What is the maximum height reached by the stone? 2
- (b) Find the coefficient of x^7 in the expansion of $\left(x^2 - \frac{1}{x}\right)^{12} \left(5 - \frac{1}{x^2}\right)^6$. 3
- (c) Find the Cartesian equation of a curve with the parametric equations 2
 $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$.

Question 7 (12 marks) Use a SEPARATE sheet of paper.

Marks

(a) Let $(3+2x)^{20} = \sum_{r=0}^{20} a_r x^r$

(i) Write down an expression for a_r .

1

(ii) Show that $\frac{a_{r+1}}{a_r} = \frac{40-2r}{3r+3}$.

1

(iii) Hence, or otherwise, find the value of the greatest coefficient in the expansion of $(3+2x)^{20}$.

2

(b) Consider the function $f(x) = \frac{1}{1+x^2}$,

(i) Sketch the function $y = f(x)$, finding any asymptotes and stationary points.

2

(ii) Write down the largest domain that contains $x = -1$ for which $y = f(x)$ has an inverse function.

1

(iii) Find the inverse function $f^{-1}(x)$ for this domain and state the domain of $f^{-1}(x)$.

2

(iv) Find the area bounded by the curve $f(x) = \frac{1}{1+x^2}$, the x axis and the values $x = -1$ and $x = 1$.

2

(v) Prove that the area between this curve and the x axis is always less than π units².

1

End of Examination

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2008 HGHS Extension 1 Trial Sol^{ns} $\sqrt{3}$

QUESTION 1

a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

$$= \lim_{x \rightarrow 0} \frac{3}{2} \cdot \frac{\sin 3x}{3x}$$

$$= \frac{3}{2}$$

b) $y = 2x - 9$
 $y = \frac{1}{3}x + \frac{9}{2}$

$$\tan \theta = \frac{2 - \frac{1}{3}}{1 + 2 \times \frac{1}{3}}$$

$$= \frac{5}{3} \times \frac{3}{5}$$

$$= 1$$

$$\theta = 45^\circ$$

c) $\frac{d(3^x)}{dx} = \frac{d e^{x \ln 3}}{dx}$

$$= \ln 3 e^{x \ln 3}$$

$$= 3^x \ln 3$$

d) $D: -\frac{1}{3} \leq x \leq \frac{1}{3}$

$$R: 0 \leq y \leq 2\pi$$

e) $I = \int_{\pi/6}^{\pi/3} \frac{dx}{\cos^2 x \tan x}$
 $= \int_{\pi/6}^{\pi/3} \frac{\sec^2 x dx}{\tan x}$

let $u = \tan x$

$$du = \sec^2 x dx$$

when $x = \pi/6$, $u = \frac{1}{\sqrt{3}}$

$x = \pi/3$, $u = \sqrt{3}$

$$\therefore I = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{du}{u}$$

$$= [\ln u]_{1/\sqrt{3}}^{\sqrt{3}}$$

$$= \ln \sqrt{3} + \ln \sqrt{3}$$

$$= 2 \ln \sqrt{3}$$

$$= \ln 3$$

$$= 1.099$$

f) i) $y = x \cos^{-1} x - \sqrt{1-x^2}$
 $\frac{dy}{dx} = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{1}{2}(1-x^2)^{-1/2} \cdot 2x$

$$= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$$

$$= \cos^{-1} x$$

ii) $\int_0^1 \cos^{-1} x dx = [x \cos^{-1} x - \sqrt{1-x^2}]_0^1$

$$= \cos^{-1}(1) - 0 - 0 + 1$$

$$= 1$$

QUESTION 2

a) $x-5 < \frac{14}{x}$

$$x^2(x-5) < 14x$$

$$x^2(x-5) - 14x < 0$$

$$x(x^2 - 5x - 14) < 0$$

$$x(x-7)(x+2) < 0$$

$$\therefore x < -2, 0 < x < 7$$

e) $\angle APQ = \angle PRQ$ (L in alt segment)
 $\angle PRQ = \angle PQR$ (k's opp equal sides)

$$\therefore \angle APQ = \angle PQR$$

$$\therefore RQ \parallel BA \text{ (alt } \angle\text{'s equal)}$$

b) $\tan 2\theta = \sqrt{3}$
 $= \tan \pi/3$

$$2\theta = n\pi + \pi/3$$

$$\theta = \frac{1}{2}n\pi + \pi/6$$

c) $f(x) = 2x^3 + ax^2 + bx + 6$

$$f(1) = 2 + a + b + 6 = -6$$

$$a + b = -14 \text{ --- (1)}$$

$$f(-2) = -16 + 4a - 2b + 6 = 0$$

$$4a - 2b = 10 \text{ --- (2)}$$

$$4 \times (1) \Rightarrow 6b = -66$$

$$b = -11$$

$$\therefore a = -3$$

$$\therefore a = -3, b = -11$$

d) $\int_0^{\pi/4} \cos^2(\frac{x}{2}) dx = \frac{1}{2} \int_0^{\pi/4} (\cos x + 1) dx$
 $= \frac{1}{2} [\sin x + x]_0^{\pi/4}$
 $= \frac{1}{2} \left(\frac{1}{\sqrt{2}} + \frac{\pi}{4} \right)$

$$= \frac{1}{2\sqrt{2}} + \frac{\pi}{8} = \frac{2\sqrt{2} + \pi}{8}$$

QUESTION 3.

a) i) $7' = 5040$
 ii) $7' - 6' \times 2 = 3600$

b) i) $f(x) = x^3 + ax^2 + bx + c$
 $f'(x) = 3x^2 + 2ax + b$
 $f''(x) = 6x + 2a$

now $3\lambda^2 + 2a\lambda + b = 0$
 and $3\beta^2 + 2a\beta + b = 0$

$\therefore 3\lambda^2 + 2a\lambda = 3\beta^2 + 2a\beta$
 $\lambda^2 - \beta^2 = \frac{2}{3}a\beta - \frac{2}{3}a\lambda$
 $(\lambda - \beta)(\lambda + \beta) = -\frac{2}{3}a(\lambda - \beta)$

$\therefore \lambda + \beta = -\frac{2}{3}a$

ii) pt of inflexion when $f''(x) = 0$

$f''(x) = 6x + 2a$
 $= 6\left(\frac{\lambda + \beta}{2}\right) + 2a$
 $= 3\left(-\frac{2}{3}a\right) + 2a$
 $= -2a + 2a$
 $= 0$

c) i) $T = 20 + Ae^{-kt}$
 $\frac{dT}{dt} = -kAe^{-kt}$
 $= -k(T - 20)$

ii) when $t = 0, T = 80$

$\therefore 80 = 20 + A$

$A = 60$

when $t = 10, T = 50$

$\therefore 50 = 20 + 60e^{-10k}$

$60e^{-10k} = 30$

$e^{-10k} = 0.5$

$-10k = \ln 0.5$

$k = 0.0693$

iii) $T = 20 + 60e^{-0.0693t}$

when $T = 30$

$30 = 20 + 60e^{-0.0693t}$

$10 = 60e^{-0.0693t}$

$e^{-0.0693t} = \frac{1}{6}$

$-0.0693t = \ln\left(\frac{1}{6}\right)$

$t = 25.86 \text{ min}$

$= 26 \text{ min}$

or $\lambda + \beta$ are both roots
 of $f'(x)$

$\therefore \lambda + \beta = -\frac{2a}{3}$ sum of
 the roots.

QUESTION 4.

a) i) $x = R \cos(nt + \alpha) + 1$

$T = \frac{2\pi}{n}$

$\pi = \frac{2\pi}{n}$

$\therefore n = 2$

$\therefore x = R \cos(2t + \alpha) + 1$

$\frac{dx}{dt} = -2R \sin(2t + \alpha)$

when $t = 0, \frac{dx}{dt} = 0$

$\therefore 0 = -2R \sin \alpha$

$\alpha = 0$

$\therefore x = R \cos(2t) + 1$

when $t = 0, x = -3$

$-3 = R + 1$

$R = -4$

$\therefore R = -4$

$n = 2$

$\alpha = 0$

ii) $f'(x) = \ln x + 1$

$f'(2) = \ln 2 + 1$

$= 1.693$

$x = 2 - 0.386$

1.693

$= 1.772$

$\therefore x = 1.8$

c) L.H.S. = $\frac{2 \tan \theta}{1 + \tan^2 \theta}$

$= \frac{2 \sin \theta}{\cos \theta} \times \cos^2 \theta$

$= 2 \sin \theta \cos \theta$

$= \sin 2\theta$

$= \sin 2\theta$

$= R.H.S.$

d) i) ${}^{25}C_2 \left(\frac{1}{9}\right)^2 \left(\frac{8}{9}\right)^{23} = 0.2467$

ii) $V = 8 \sin\left(2 \times \frac{\pi}{6}\right)$

$= 8 \sqrt{3}$

$= 4\sqrt{3}^2$

ii) $1 - {}^{25}C_0 \left(\frac{8}{9}\right)^{25} - {}^{25}C_1 \left(\frac{1}{9}\right)^1 \left(\frac{8}{9}\right)^{24}$

$= 1 - 0.05262 - 0.16445$

$= 0.7829$

b) Let $f(x) = x \ln x - 1$

$f(1) = 1 \ln 1 - 1$

$= -1$

$f(2) = 2 \ln 2 - 1$

$= 0.4$

$\therefore$ soln lies between $x = 1, 2$.

QUESTION 5

a) • Prove true for $n=1$

$$LHS = 1^3 = 1$$

$$RHS = 1^2 = 1$$

∴ True for $n=1$

• Assume true for $n=k$

$$\text{i.e. } 1^3 + 2^3 + \dots + k^3 = (1+2+3+\dots+k)^2$$

• Prove true for $n=k+1$

$$\text{i.e. } 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = (1+2+\dots+k+(k+1))^2$$

$$L.H.S. = (1+2+\dots+k)^2 + (k+1)^3$$

$$= \left[\frac{k(k+1)}{2} \right]^2 + (k+1)^3$$

$$= (k+1)^2 \left(\frac{k^2}{4} + k+1 \right)$$

$$= (k+1)^2 \left(\frac{k^2 + 4k + 4}{4} \right)$$

$$= (k+1)^2 \left(\frac{(k+2)^2}{4} \right)$$

$$R.H.S. = \left[\frac{k+1+(k+1+1)}{2} \right]^2$$

$$= (k+1)^2 \left(\frac{(k+2)^2}{4} \right)$$

$$= L.H.S.$$

∴ True for $n=k+1$

Sum of AP
k terms

Sum of AP
k+1 terms

QUESTION 5

b) i) $\frac{r}{3} = \frac{h}{10}$ similar Δ 's

$$r = \frac{3h}{10}$$

$$\text{ii) } \frac{dv}{dt} = 2$$

$$V = \frac{\pi r^2 h}{3}$$

$$\frac{dV}{dh} = \frac{\pi \cdot 9h^2 \cdot h}{300}$$

$$= \frac{9\pi h^3}{300}$$

$$\frac{dV}{dh} = \frac{27\pi h^2}{300}$$

$$\therefore \frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{100}{9\pi h^2} \times 2$$

$$= \frac{200}{9\pi h^2}$$

when $h=8$

$$\frac{dh}{dt} = \frac{100}{\pi \times 8^2}$$

$$= \frac{0.1105}{\pi}$$

$$= \frac{25}{497} \text{ m/min.}$$

$$= \frac{25}{22\pi}$$

c) Let roots be $\frac{a}{r}, a, ar$

$$\frac{a}{r} + a + ar = 21$$

$$a^3 = 216$$

$$a = 6$$

$$\therefore \frac{1}{r} + 1 + r = \frac{7}{2}$$

$$7r = 2 + 2r + 2r^2$$

$$2r^2 - 5r + 2 = 0$$

$$(2r-1)(r-2) = 0$$

$$\therefore r = \frac{1}{2}, 2$$

∴ roots are, 3, 6, 12

$$\text{d) } \int_0^{0.4} \sin^{-1} x \, dx = 0.4 \left[0 + 4 \times 0.201 + 0.4 \right]$$

$$= 0.081$$

$$= 0.1$$

QUESTION 6

$$a) i) \ddot{y} = -10$$

$$\dot{y} = -10t + 10 \sin \theta$$

$$= -10t + \frac{10 \times 3}{5}$$

$$= -10t + 6$$

$$y = \frac{-10t^2}{2} + 6t + 27$$

$$= -5t^2 + 6t + 27$$

ii) when $y = 0$

$$5t^2 - 6t - 27 = 0$$

$$(5t + 9)(t - 3) = 0$$

$$\therefore t = 3$$

$$\text{when } t = 3, x = 8 \times 3$$

$$= 24$$

iii) max height when $\dot{y} = 0$

$$\text{i.e. } -10t + 6 = 0$$

$$t = \frac{6}{10}$$

$$\text{when } t = 0.6, y = -5(0.6)^2 + 6(0.6) + 27$$

$$= 28.8$$

$$b) \sum_{k=0}^{12} {}^{12}C_k (x^2)^{12-k} (-1)^k (x^{-1})^k \cdot \sum_{r=0}^6 {}^6C_r 5^{6-r} (-1)^r (x^{-2})^r$$

$$\therefore x^{2k-2k-k-2r} = x^{24-3k-2r}$$

$$\therefore 24 - 3k - 2r = 7$$

$$3k + 2r = 17$$

k	r	3k+2r
5	1	17
3	4	17

$$\therefore \text{Coeffs} = {}^{12}C_5 (-1)^5 \cdot {}^6C_1 5^5 (-1)^1$$

$$+ {}^{12}C_3 (-1)^3 \cdot {}^6C_4 5^2 (-1)^4$$

$$= 14850000 - 82500$$

$$= 14767500$$

QUESTION 7

$$a) i) (3+2x)^{20} = \sum_{r=0}^{20} {}^{20}C_r 3^{20-r} \cdot 2^r x^r$$

$$\therefore a_r = {}^{20}C_r \cdot 3^{20-r} \cdot 2^r$$

$$ii) \frac{a_{r+1}}{a_r} = \frac{{}^{20}C_{r+1} \cdot 3^{19-r} \cdot 2^{r+1}}{{}^{20}C_r \cdot 3^{20-r} \cdot 2^r}$$

$$= \frac{20!}{(r+1)!(19-r)!} \times \frac{r!(20-r)!}{20!} \times \frac{2}{3}$$

$$= \frac{20-r}{r+1} \times \frac{2}{3}$$

$$= \frac{40-2r}{3r+3}$$

$$iii) \frac{40-2r}{3r+3} \geq 1$$

$$40-2r \geq 3r+3$$

$$5r \leq 37$$

$$r \leq 7\frac{4}{5}$$

$$\therefore r = 7$$

$$\therefore a_{r+1} = a_8 = {}^{20}C_8 \cdot 3^{12} \cdot 2^8$$

$$= 1.71 \times 10^{13}$$

QUESTION 7.

$$b) i) f(x) = \frac{1}{1+x^2}$$

$$f'(x) = -(1+x^2)^{-2} \cdot 2x$$

$$= \frac{-2x}{(1+x^2)^2}$$

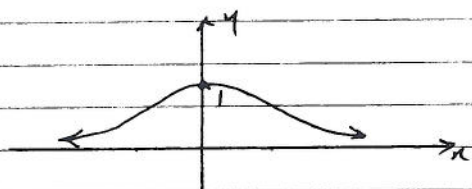
$$\lim_{x \rightarrow \infty} \frac{1}{1+x^2} = 0^+$$

$$\lim_{x \rightarrow -\infty} \frac{1}{1+x^2} = 0^+$$

$$\text{stat pt when } -2x = 0$$

$$x = 0$$

$$\text{when } x = 0, y = 1.$$



$$ii) x \leq 0$$

$$iii) x = \frac{1}{1+y^2}$$

$$1+y^2 = \frac{1}{x}$$

$$y^2 = \frac{1}{x} - 1$$

$$y = -\sqrt{\frac{1-x}{x}}$$

$$\text{Domain } 0 < x \leq 1$$

$$iv) A = \int_{-1}^1 \frac{1}{1+x^2} dx$$

$$= [\tan^{-1} x]_{-1}^1$$

$$= \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \pi$$

$$v) A \leq 2 \lim_{a \rightarrow \infty} \int_0^a \frac{1}{1+x^2} dx$$

$$\leq 2 \lim_{a \rightarrow \infty} [\tan^{-1} x]_0^a$$

$$\leq 2 \lim_{a \rightarrow \infty} (\tan^{-1} a - 0)$$

$$\leq 2 \times \frac{\pi}{2}$$

$$\leq \pi.$$

$\therefore$ area always less than π .